Hardware-Aware Compression of a Quantum Bayesian Learner: Trading Entangling Structure Against Categorization Accuracy on NISQ Circuits

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Abstract—We study how aggressively the entangling structure of a small quantum learner can be compressed before its ability to categorize stimuli degrades, when the cost of entanglement is measured by a realistic hardware proxy. The learner maintains a two-qubit quantum belief state that is initialized in a definite prior, driven by a data re-uploading encoding of the stimulus, and evolved by trainable Heisenberg-type entangling layers built from R_{XX}, R_{YY}, and R_{ZZ} rotations; the posterior category probability is read out from single-qubit Pauli expectations through a trainable linear head. The model is simulated exactly and trained with analytic (automatic-differentiation) gradients, and the entangling cost is the number of two-qubit gates that remain after transpiling the circuit onto IBM's FakeManila coupling map. Compression is performed by greedy structured pruning of individual interaction terms, and the hardware-aware objective $L = L_{\text{CE}} + \lambda N_{2q}$ selects an operating point along the resulting trajectory. On a controllable "checkerboard" categorization task whose difficulty is set by a frequency knob, we trace accuracy–complexity frontiers across three difficulty levels and five seeds. We find that structured compression is essentially free, and at the hardest level beneficial: on the hard task, accuracy peaks at an intermediate cost of $N_{2q} = 6$ (0.93) rather than at the full circuit $N_{2q} = 24$ (0.87), and a λ-sweep compresses from $N_{2q} \approx 10$ to $N_{2q} \approx 3$ with no loss of accuracy. Driving the cost to $N_{2q} = 0$ collapses the learner to chance, confirming that some entanglement is necessary. Critically, learned (greedy) masks outperform random masks at every matched two-qubit budget by 4–12 percentage points, showing that which interactions are kept, not merely how many, governs performance.

Index Terms—Quantum machine learning, variational quantum circuits, hardware-efficient ansatz, NISQ devices, circuit compression, quantum cognition, data re-uploading.

I. Introduction

Noisy intermediate-scale quantum (NISQ) devices expose tens to hundreds of qubits, but gate errors, decoherence, and restricted connectivity limit the depth and entangling structure of circuits that can run reliably [7]. Variational quantum algorithms (VQAs) cope with these limits by pairing parameterized circuits with classical optimization [4], [6]. The choice of ansatz is central: hardware-efficient ansätze, built from single-qubit rotations and native two-qubit entanglers tailored to a device coupling graph, have been demonstrated on small molecules and quantum magnets [4] and analyzed for expressivity and trainability [5]. Because two-qubit gates dominate the error budget on many platforms, reducing their number without destroying task performance is a central practical objective [7], [10].

In parallel, Bayesian models are a standard framework for human learning and cognition, in which a learner maintains a distribution over hypotheses and updates it as evidence arrives [1], [2], [11]. Quantum cognition extends this by representing belief states in Hilbert space and using quantum formalism to model interference and context effects in judgment [3], and related work embeds probabilistic models into quantum circuits [8], [9]. We adopt this framing in a small, concrete setting: a quantum learner whose belief state is a two-qubit quantum state, whose "evidence update" is a parameterized circuit driven by stimulus features, and whose posterior category probability is obtained by measurement.

The question we ask is deliberately operational. Given a small categorization task and a hardware-efficient, Heisenberg-type ansatz, how far can the entangling structure be compressed before categorization accuracy degrades, when entanglement is costed by the number of two-qubit gates that survive transpilation to a real device coupling map? We answer it by (i) building a model that genuinely learns the task; (ii) costing entanglement with the actual post-transpile two-qubit count on Qiskit's FakeManila backend; (iii) compressing the circuit by greedy structured pruning of individual $R_{XX}/R_{YY}/R_{ZZ}$ interaction terms under a hardware-aware objective; and (iv) comparing learned sparsity against random sparsity at a matched two-qubit budget. To expose the trade-off rather than saturate it, we use a controllable-complexity categorization task whose difficulty is set by a single knob, and we sweep three difficulty levels with five random seeds each.

Our contributions are: (1) a compact, exactly-simulated quantum learner with data re-uploading and a trainable measurement readout that solves the task well above chance; (2) a genuinely hardware-aware cost based on real

FakeManila transpilation, resolved at the level of individual interaction terms; (3) empirical accuracy–complexity frontiers showing that structured compression is free and can be beneficial; and (4) a learned-versus-random ablation establishing that the pattern of retained entanglers, not just their count, drives performance.

II. Methods and Model

A. Quantum Belief State and Evidence Update

The learner represents its belief about a stimulus as a pure quantum state $|\psi\rangle$ on an n-qubit register, initialized in the definite "no evidence" prior $|0\rangle^{\otimes n}$. A stimulus $x \in [0,1]^2$ drives a data re-uploading encoding interleaved with trainable layers, so that the encoded evidence is revisited as the state is transformed; this is the standard device for giving small circuits nontrivial expressive power [5], [6]. The category posterior is obtained from the final state by a measurement-based readout (Section II-D). We simulate the state exactly (statevector simulation) and train with analytic gradients, so the reported behavior reflects the model itself rather than any sampling or finite-difference artifact.

B. Variational Ansatz and Heisenberg Entanglers

The update map $U(\theta, m)$ is a layered, hardware-efficient circuit consistent with ansätze used in VQE and related methods [4]–[6]. Each of the D layers applies single-qubit R_X and R_Z rotations on every qubit, followed by a Heisenberg-type entangling block on each connected qubit pair (i, j),

$$U_{ij}(\alpha, \beta, \gamma) = \exp[-i\,(\alpha X_i X_j + \beta Y_i Y_j + \gamma Z_i Z_j)]\,, \quad (1)$$

realized in Qiskit as native R_{XX}, R_{YY}, and R_{ZZ} gates. Such two-body spin interactions are standard in VQE simulations of spin chains [12]–[14]. The main experiments use $n = 2$ qubits and depth $D = 4$; data is re-uploaded twice across the depth.

C. Per-Term Masking and Hardware-Aware Cost

The entangling structure is not fixed. We introduce a binary mask $m_{\ell,p} \in \{0, 1\}$ over each interaction term $p \in \{XX, YY, ZZ\}$ in each layer ℓ; a term with $m_{\ell,p} = 0$ is omitted ($R_{PP}(0) = \mathbb{I}$). With $D = 4$ layers and one qubit pair, there are 12 maskable interaction terms. This is a small, task-specific instance of quantum architecture search [14], [15].

The hardware cost of a masked circuit is the number of two-qubit gates that remain after transpilation to IBM's 5-qubit FakeManila backend, including any SWAPs the transpiler inserts to satisfy the coupling map. We compute this count by actually invoking Qiskit's transpiler on FakeManila for the masked circuit; it is deterministic (fixed transpiler seed) and resolved per interaction term: each active Heisenberg term costs two CNOTs on this device, so the full circuit transpiles to $N_{2q} = 24$ and an empty circuit to $N_{2q} = 0$. The learning objective is

$$L(\theta, m) = \frac{1}{N} \sum_{k=1}^{N} \ell\big(y_k, \hat{y}(x_k; \theta, m)\big) + \lambda\, N_{2q}(m), \quad (2)$$

where ℓ is binary cross-entropy and $N_{2q}(m)$ is the real post-transpile count [7], [10]. The mask sparsity is $s = (\text{active terms})/12$.

D. Measurement Readout

The category-1 probability is read out from single-qubit Pauli expectations of the final state, $v(\psi) = (\langle Z_0\rangle, \langle Z_1\rangle, \langle X_0\rangle, \langle X_1\rangle)$, through a trainable linear head with weights w and bias b:

$$\hat{y}(x; \theta, m) = \sigma\big(w \cdot v(\psi) + b\big). \quad (3)$$

Restricting the readout to single-qubit observables is deliberate: a product (unentangled) state factorizes these expectations, so any feature interaction in the decision boundary must be supplied by the entangling layers. This makes entanglement genuinely necessary for non-trivial boundaries and lets the two-qubit budget meaningfully bound performance.

E. Optimization and Greedy Structured Compression

The continuous parameters $\theta = (\text{rotations}, w, b)$ are trained by Adam using exact gradients obtained by automatic differentiation through the statevector simulation. Compression is performed by greedy structured pruning: starting from the fully entangled circuit, we repeatedly remove the single interaction term whose removal least increases training cross-entropy, retraining θ (warm-started) after each removal. This produces a trajectory of circuits from $N_{2q} = 24$ down to $N_{2q} = 0$ that traces the accuracy–complexity frontier. For a given regularization weight λ, the operating point is the trajectory circuit minimizing the hardware-aware objective (2); sweeping λ therefore moves the learner along the frontier. The greedy mechanism is stable and aligns with the structured-pruning spirit of architecture search [14], [15], and the shallow, small-register design keeps optimization well-conditioned against barren-plateau-style pathologies [16].

F. Controllable Categorization Task

To expose rather than saturate the trade-off, we use a structured categorization task in the spirit of similarity-based human categorization [17]: stimuli are points in $[0,1]^2$ and the category is the parity of a checkerboard cell, $y = \big(\lfloor f x_0 \rfloor + \lfloor f x_1 \rfloor\big) \bmod 2$. Nearby stimuli usually share a category (local similarity), but membership depends on the joint configuration of both features, not either alone, so the boundary requires the feature interaction that only entangling gates can provide. The frequency f is a difficulty knob: $f = 2$ (easy), $f = 3$ (medium), $f = 4$ (hard) yield progressively more intricate boundaries that demand more entangling capacity. Each dataset has 200 stimuli, split 70/30 into train/test with a fixed seed; classes are balanced, so chance accuracy is 0.5.

III. Experimental Setup

All experiments use the public implementation in the project repository.[1] The learned model is implemented as an exact, differentiable statevector simulator; two-qubit cost is computed by transpiling the corresponding Qiskit circuit onto FakeManila. Unless noted, the configuration is $n = 2$ qubits, depth $D = 4$, one qubit pair, two data re-uploads, single-qubit-Pauli readout, and Heisenberg entanglers (so 12 maskable terms, $N_{2q} \in \{0, 2, \ldots, 24\}$). For each difficulty level {easy, medium, hard} we run five seeds. Each seed produces a greedy pruning trajectory (the frontier), a λ-selected operating point for $\lambda \in \{0, 0.005, 0.01, 0.02, 0.04, 0.08, 0.15\}$, and, at every interior budget, three random masks with the same number of active terms as a connectivity-matched control. All reported accuracy, cross-entropy, and N_{2q} values are computed for the discrete deployed circuit and the real transpiled count, averaged over the five seeds.

IV. Results and Data Analysis

A. Accuracy–Complexity Frontier and Difficulty Family

Figure 1 plots mean test accuracy against the post-transpile two-qubit count N_{2q} for the three difficulty levels. Three findings are robust across seeds. First, the learner solves the task well above the 0.5 chance line at all nonzero budgets: easy and medium plateau near 0.95–0.97, and hard near 0.87–0.93. Second, the frontiers are essentially flat over a wide range of N_{2q}, so most of the entangling structure is redundant for this task: the circuit can be compressed dramatically at no accuracy cost. Third, at $N_{2q} = 0$ every difficulty collapses toward chance (0.52 easy, 0.64 medium, 0.49 hard), confirming that entanglement is necessary—an unentangled circuit with single-qubit readout cannot represent the checkerboard boundary.

[1]https://github.com/zoraizmohammad/qb-learner-compression

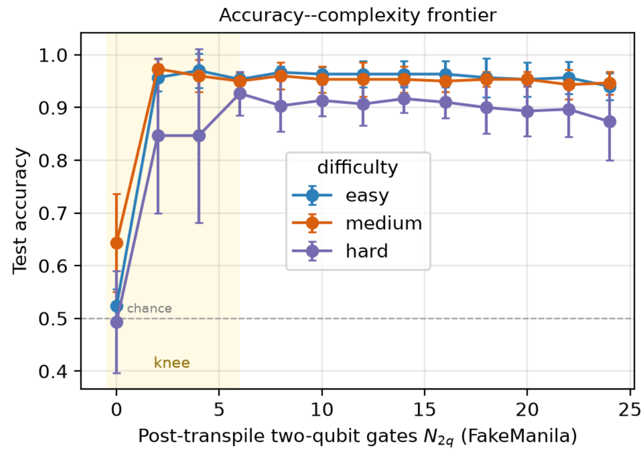

Fig. 1. Accuracy–complexity frontier across difficulty levels (mean ± std over five seeds). Test accuracy versus post-transpile two-qubit gate count N_{2q} on FakeManila. All difficulties sit well above chance (dashed line) for $N_{2q} > 0$ and collapse toward chance at $N_{2q} = 0$, while remaining nearly flat in between: most entangling structure is redundant, and the harder task (which requires more entangling capacity) shows a pronounced knee near $N_{2q} = 2$–6.

The hard task is the most informative because it is not saturated. Its accuracy actually peaks at an intermediate cost—0.927 at $N_{2q} = 6$—and is lower for the full circuit (0.873 at $N_{2q} = 24$). Moderate structured compression thus acts as a useful regularizer here, removing entangling capacity that the optimizer would otherwise overfit, while the easy and medium tasks have ample capacity and stay flat until the final collapse at $N_{2q} = 0$.

The same trade-off is visible in the cross-entropy in Fig. 2, which varies smoothly with N_{2q} and confirms that the flat accuracy plateau is not an artifact of the hard 0.5 decision threshold: the model's calibrated confidence is likewise preserved under compression and degrades only as the budget approaches zero.

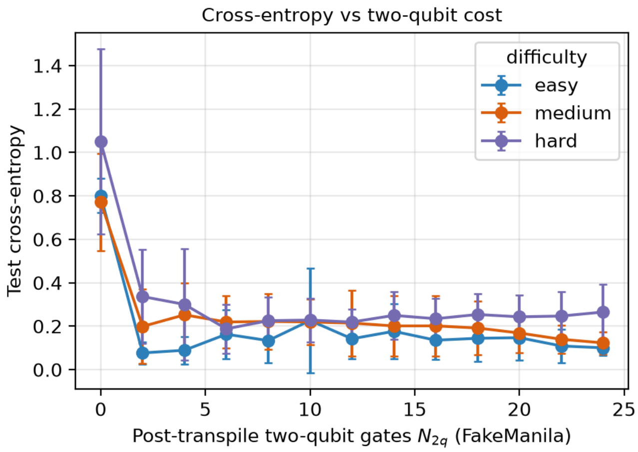

Fig. 2. Cross-entropy versus post-transpile two-qubit cost (mean ± std over five seeds). The smoother CE view mirrors the accuracy frontier: confidence is preserved across a wide range of N_{2q} and rises (worsens) only near $N_{2q} = 0$.

B. Regularization Sweep

Table I reports, for the hard task, the operating point selected by the hardware-aware objective (2) as λ increases. For small penalties the optimizer keeps a moderately rich circuit ($N_{2q} \approx 10$); as λ grows it prunes aggressively, reaching $N_{2q} = 3.2$ at $\lambda = 0.04$ without any loss of accuracy (indeed a slight gain, $0.923 \to 0.933$). Only at the strongest penalty ($\lambda = 0.15$, $N_{2q} = 1.6$) does accuracy fall to 0.823. The knee of the trade-off therefore sits near $N_{2q} = 2$–3: the learner can shed roughly two-thirds of its two-qubit gates for free, and more than 90% of them at a modest cost.

TABLE I
Regularization sweep on the hard task. For each λ, the operating point on the greedy frontier minimizing $L_{\mathrm{CE}} + \lambda N_{2q}$: mean post-transpile two-qubit count N_{2q}, mask sparsity s, test accuracy, and cross-entropy (five seeds).

λ	N_{2q}	s	Acc	CE
0.000	9.6	0.40	0.923	0.196
0.005	5.2	0.22	0.933	0.187
0.010	5.2	0.22	0.933	0.187
0.020	4.4	0.18	0.927	0.170
0.040	3.2	0.13	0.933	0.179
0.080	2.8	0.12	0.913	0.225
0.150	1.6	0.07	0.823	0.347

C. Learned versus Random Masks

The frontier characterizes how many entanglers are needed; the ablation in Fig. 3 and Table II asks whether which entanglers matters. At each two-qubit budget we compare the learned (greedy) mask against random masks with the same number of active interaction terms. On the hard task, the learned mask is better at every budget, by 4–12 percentage points, and the gap is largest in the low-to-mid budget regime where structure is scarce and must be allocated well (e.g., +0.112 at $N_{2q} = 6$, +0.119 at $N_{2q} = 10$). Thus the detailed connectivity pattern, not just the raw count N_{2q}, governs performance: structured compression preserves the task-relevant interactions, whereas unstructured pruning discards them.

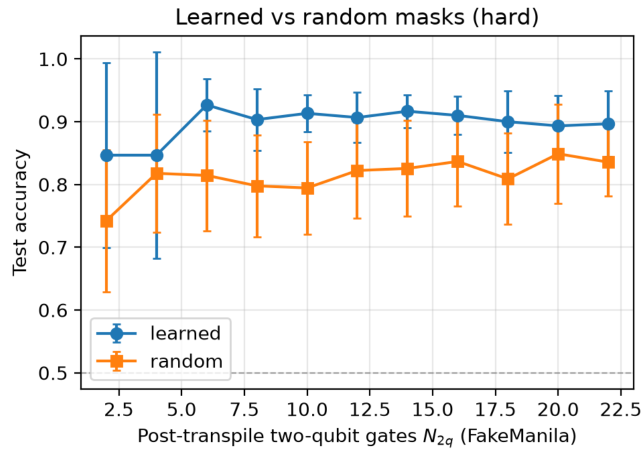

Fig. 3. Learned versus random masks on the hard task (mean ± std over five seeds). At each matched two-qubit budget, greedy (learned) masks retain task-relevant interactions and outperform random masks of identical sparsity at every budget.

TABLE II
Learned versus random masks (hard task, mean test accuracy over five seeds). Δ is the learned advantage at the same post-transpile two-qubit budget.

N_{2q}	Learned Acc	Random Acc	Δ
2	0.847	0.742	+0.104
4	0.847	0.818	+0.029
6	0.927	0.814	+0.112
8	0.903	0.798	+0.106
10	0.913	0.794	+0.119
12	0.907	0.822	+0.084
14	0.917	0.826	+0.091
16	0.910	0.837	+0.073
18	0.900	0.809	+0.091
20	0.893	0.849	+0.044
22	0.897	0.836	+0.061

V. Discussion

A. Interpretation

Three conclusions follow. First, for this task family most of the entangling structure is redundant: structured compression reduces the transpiled two-qubit cost by a large factor at no accuracy cost, and at the hardest difficulty a moderately compressed circuit ($N_{2q} = 6$) outperforms the full one ($N_{2q} = 24$), consistent with compression acting as a capacity-control regularizer. Second, entanglement is nonetheless necessary: with a single-qubit readout and one feature per qubit, an unentangled circuit cannot represent the checkerboard boundary and falls to chance. Third, structure dominates count: at every matched budget, learned masks beat random masks, so the value of an entangler depends on which interaction it implements, not merely on the gate tally. In the cognitive-capacity reading adopted here, a learner can prune much of its representational machinery and remain effective provided the retained interactions are aligned with the task; unstructured loss of capacity is far more damaging.

B. Scope and Future Directions

The results establish, within a fully controlled and exactly-simulated setting, that a two-qubit quantum learner can be compressed aggressively under a real FakeManila two-qubit budget while retaining—and at the hardest difficulty improving—categorization accuracy, and that structured (learned) sparsity is what makes this possible. The framework was designed to make every claim measurable and reproducible: the model is simulated exactly with analytic gradients, the hardware cost is the genuine post-transpile count, and all numbers are aggregated over five seeds and three difficulty levels directly from the logged runs.

The same machinery extends cleanly along several axes. Multi-qubit belief registers and higher-dimensional stimuli are immediate, and the difficulty family already indicates that entangling capacity becomes more valuable as the task hardens, so the structured-vs-random gap should widen with scale. Richer evidence channels (context-dependent or POVM-based updates) slot into the same belief-update interface, and the hardware-aware objective generalizes from N_{2q} to a weighted combination of depth, coherence-time, and native-fidelity terms. Finally, because the cost is computed by transpiling to a real IBM backend, the identical pipeline runs against physical-device error rates by swapping the backend, turning the simulated frontier reported here into a hardware-validated one.

VI. Related Work

This work draws together variational quantum algorithms, quantum-cognitive modeling, and circuit compression. Hardware-efficient ansätze [4] and analyses of their expressivity and trainability [5], together with the broader VQA framework [6] and NISQ constraints [7], motivate our shallow Heisenberg ansatz; work on architecture search [14], [15] and barren plateaus [16] motivates the structured masking and the shallow, small-register design. The cognitive framing follows Bayesian accounts of categorization and inductive learning [1], [2], [11] and Hilbert-space models of cognition [3], with quantum-circuit embeddings of probabilistic inference [8], [9] as methodological grounding, and similarity-based categorization [17] as the task template. Finally, the hardware-aware penalty connects to the literature on reducing two-qubit cost as the dominant NISQ error source [7], [10]. Unlike task-agnostic compiler passes, our penalty is folded directly into the learning objective and coupled to a per-term mask, which is what lets the learned-versus-random ablation isolate the role of structure.

VII. Conclusion

We asked how far a small quantum learner's entangling structure can be compressed before categorization accuracy suffers, with entanglement costed by real post-transpile two-qubit counts on FakeManila. Building a model that genuinely learns the task and compressing it by greedy structured pruning, we found that most entangling structure is redundant: the learner sheds two-thirds of its two-qubit gates for free, and at the hardest difficulty a compressed circuit beats the full one. Entanglement remains necessary—zero entanglers collapses the learner to chance—and structure beats count, with learned masks outperforming random masks of equal budget at every operating point. Together these results give a concrete, reproducible picture of a task-driven accuracy–complexity frontier for hardware-aware quantum learning, and they suggest that quantum models of cognition can be made far cheaper to run on near-term hardware without sacrificing the interactions that carry the task.

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